Anveshan
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MLH MAJOR LEAGUE HACKING | unstop | Google Gemma | Reforged '26 | n8n | Hindsight

Valid8

One-tap verification against fake kidnapping and family-emergency scam calls

Team Name: Hermes

Team Members

Aryan Singhal, Anshu Gautam, Rohit Bora, Saurav Chakraborty

Track Name: N8N

IDEA

The Problem

- With just **3–4 seconds of audio** from social media, free tools can clone a voice at an estimated **~85% match** (McAfee Labs, 2023).
- Scammers call parents in that voice, claiming a **kidnapping, arrest or accident**, and demand an urgent UPI transfer.
- Their key trick is **isolation**: “phone mat kaatna”. While the parent stays on the line, they cannot check on the child.
- Under panic, parents pay within minutes, before anyone confirms the child is actually safe.

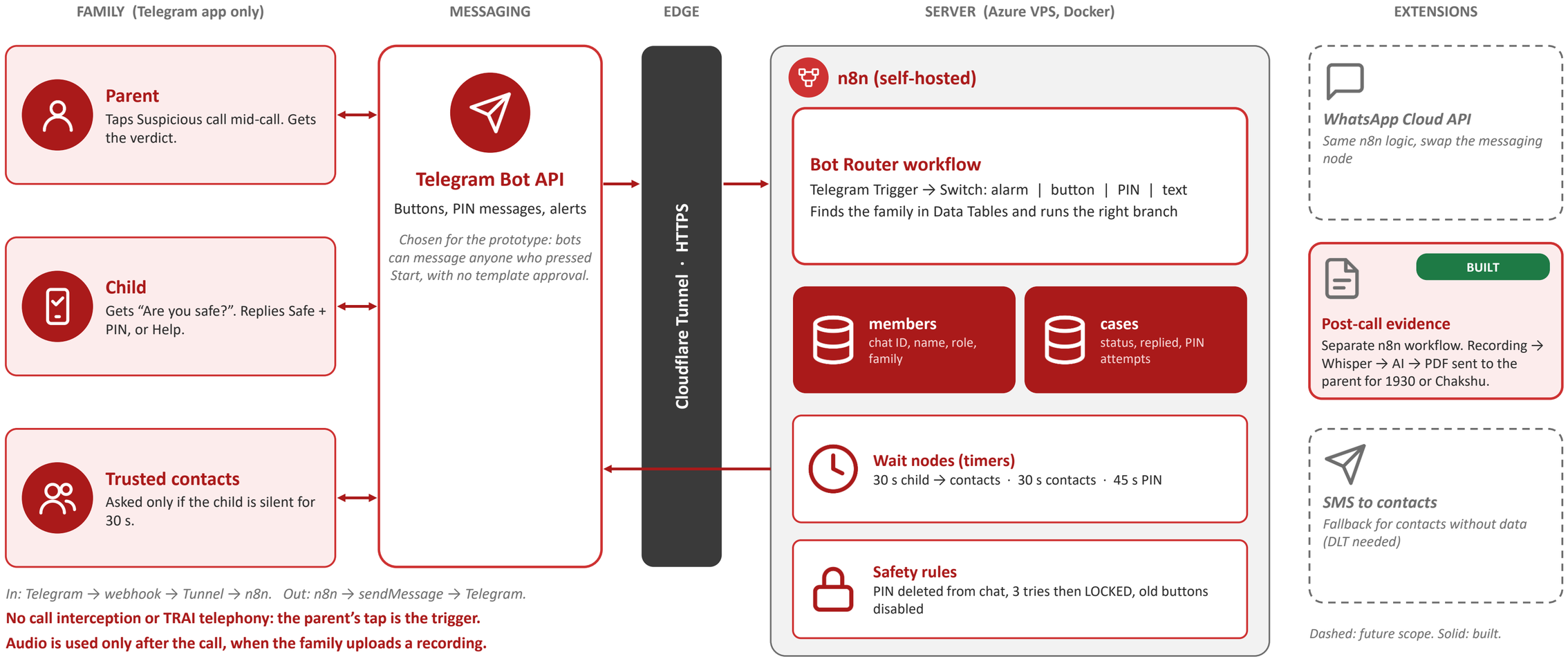
The Solution

- Valid8 is a **Telegram bot powered by n8n**. The parent stays on the call and taps **Suspicious call**.
- The child instantly gets **“Are you safe?”** and confirms with a private **Safe PIN**, or presses **Help**.
- No reply in 30 seconds? Valid8 asks the family’s **trusted contacts**.
- The parent gets a clear verdict within about a minute: **SAFE, HELP, LOCKED or UNCONFIRMED**. Valid8 never says “safe” without the child’s PIN.
- After the call, a recording becomes a **PDF evidence report** for 1930 or Chakshu.

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TECHNICAL ARCHITECTURE



In: Telegram → webhook → Tunnel → n8n. Out: n8n → sendMessage → Telegram.

No call interception or TRAI telephony: the parent’s tap is the trigger.

Audio is used only after the call, when the family uploads a recording.

Dashed: future scope. Solid: built.

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WORKFLOW



Scam call

Caller claims the child is kidnapped and says “don’t hang up”

Parent taps alarm

Taps Suspicious call on Telegram while the call stays on

Child is asked

“Are you safe?” with I’m safe and Help buttons

Safe PIN

Child sends a 4-digit PIN. Bot deletes it. 3 tries max

Trusted contacts

If the child is silent for 30 s, contacts are asked

What the parent sees (within about a minute)

SAFE

Child confirmed with the correct Safe PIN.

PAISE MAT BHEJO. High risk of scam.

HELP

Child pressed the Help button.

Call 112 now.

LOCKED

3 wrong PINs: phone may be with someone else.

Don’t pay. Call 112.

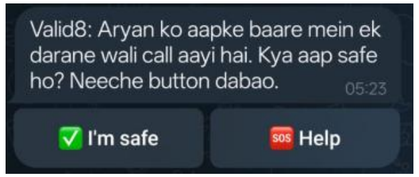
UNCONFIRMED

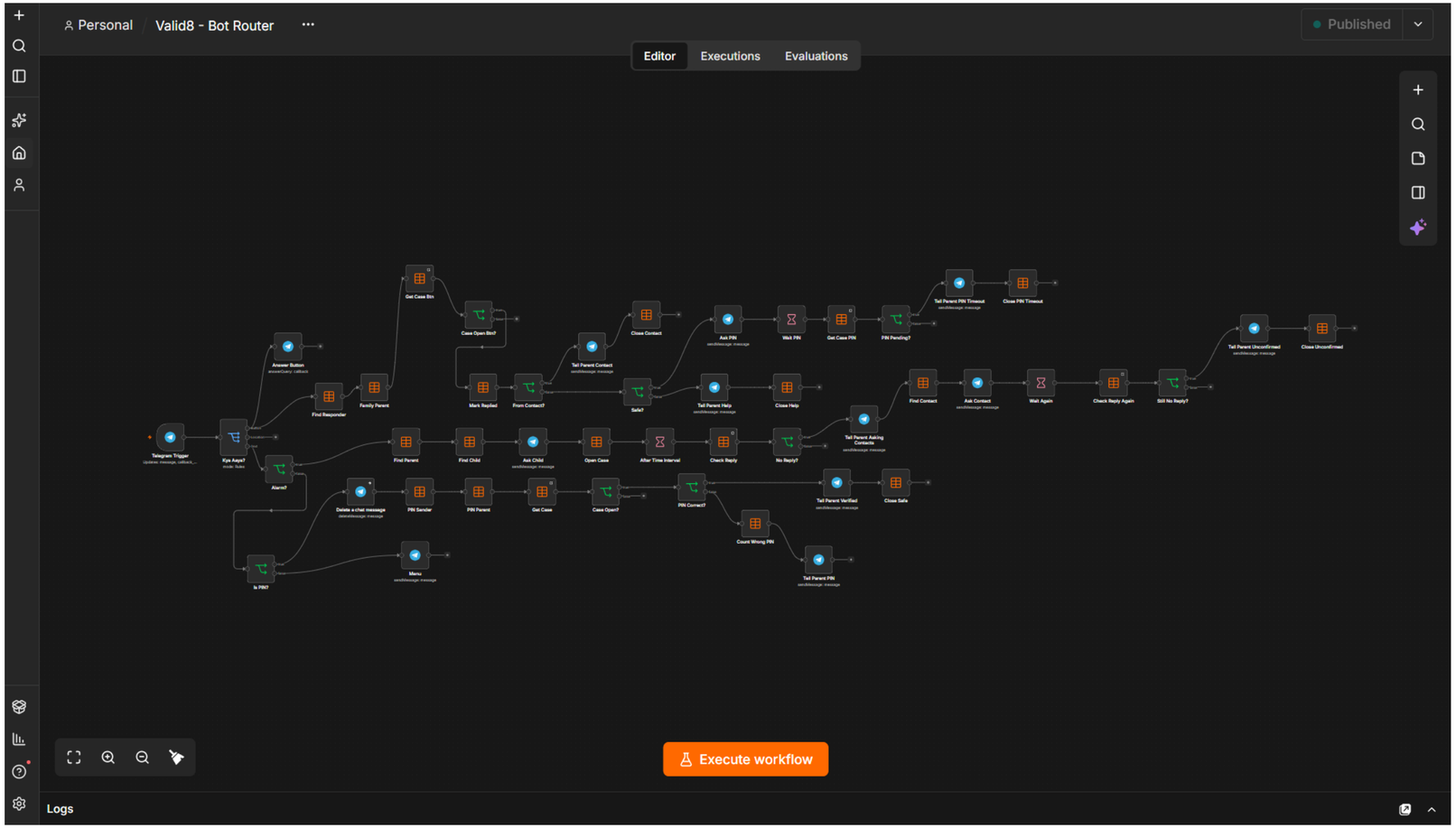
No PIN and no reply from the child or contacts.

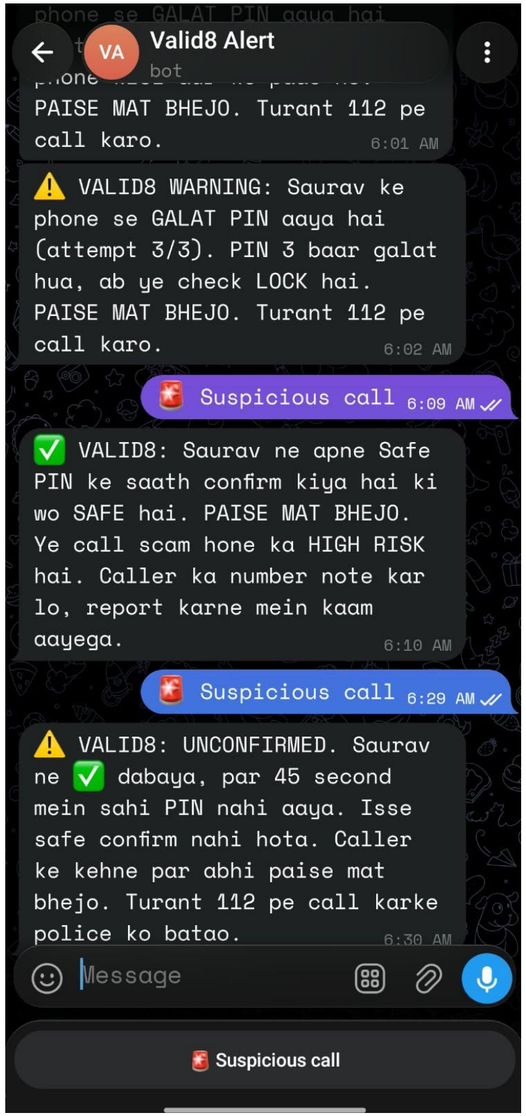
Don’t pay on the caller’s word. Call 112.

After the call: upload the recording and Valid8 sends a PDF evidence report (amount, UPI IDs, threats) for 1930 or Chakshu.

PROTOTYPE DEMO







PROTOTYPE DEMO

Part 2: post-call evidence report

Valid8 Evidence Report

Generated 28 Oct 2026, 10:03 AM (IST) for Aryan

Suspected scam: kidnap

Person named by caller	Saurav
Amount demanded	Rs 50,000
Payment method	Not mentioned
UPI IDs	Not mentioned
Phone numbers	Not mentioned
Deadline given	Not mentioned
Threats	kidnapped
Uncertain details	Not mentioned
Summary	The caller claims Saurav has been kidnapped and demands money. They ask for 50,000 INR to be sent.

Transcript (automatic, may contain errors)

> Hello! Saurav ka kidnapped kar liya gaya hai. 50000 rupaye bhejo, varno Saurav gaya kaam se.

What to do next

1. Do not send money. Keep the original recording safe.
2. If money was sent: call 1930 immediately or file at cybercrime.gov.in, and inform your bank with the UPI ID above.
3. If no money was sent: report the number on Sanchar Saathi (Chakshu).

This report was generated automatically from a call recording. Details were extracted by AI and may contain errors. Please verify every number and UPI ID against the recording before filing.

Real report generated from our test recording

1 Record or upload
After the call, a family member records it on speaker, or uploads the phone's call recording, on our web page.

2 n8n webhook
The audio reaches n8n with the family ID. Valid8 finds the parent in the members table.

3 Transcript
Whisper large-v3 (via Groq) turns Hindi and Hinglish speech into text.

4 AI extraction
An AI model pulls out the amount, UPI IDs, phone numbers, deadline and threats.

5 PDF on Telegram
Gotenberg turns it into a PDF, sent to the parent in about a minute for 1930 or Chakshu.

AI output can contain errors, so the report asks the family to verify every number and UPI ID before filing.

FEEDBACK FROM MENTORING ROUND & IMPROVED IMPLEMENTATION

WHAT OUR MENTORS SUGGESTED	HOW IT HELPED US IMPROVE VALID8
Support WhatsApp, not just Telegram	A great point, since most Indian families use WhatsApp. We studied the WhatsApp Cloud API: it needs a verified business number and approved templates, so we kept Telegram for the prototype and made the n8n logic channel-independent. WhatsApp is now our first production step.
Can you really intercept calls? (TRAI)	This question changed our design. Call interception needs TRAI-compliant telephony and raises privacy concerns, so we removed it completely. Valid8 never touches the call: the parent's one tap is the trigger. The product is simpler and safer.
Consider SMS and email	Thanks to this, we checked every channel. Email is too slow during a live call, and SMS cannot carry secure buttons or a PIN flow (and needs DLT registration). SMS is now planned as a fallback for trusted contacts without data.
Handle more types of scams	The most valuable suggestion. We realised one verify flow covers every scam that claims a loved one is in danger: fake kidnapping, fake arrest, accident or hospital, the "new number" scam, and voice-clone calls to elderly parents.

Thank you, mentors: your feedback turned Valid8 from detecting scam words in audio into verifying the child directly with a Safe PIN and trusted contacts.

USP

1	**Breaks the isolation**	Scammers say “don’t hang up”. One tap checks the child and trusted contacts while the parent stays on the line.
2	**Proof, not guesses**	The child’s own Safe PIN confirms safety. No voice-deepfake guessing, which phone compression makes unreliable.
3	**Honest verdicts**	SAFE only with the PIN. Otherwise HELP, LOCKED or UNCONFIRMED, each pointing to 112. Never “scam confirmed”.
4	**One flow, many scams**	Fake kidnapping, fake arrest, accident or hospital scams, the “new number” scam and calls to elderly parents.
5	**No app, no interception**	Works on top of any phone call with just Telegram. WhatsApp next, on the same n8n workflow.

REFERENCES AND SOURCES

Tools and platforms

n8n – Workflow automation, self-hosted
n8n.io

n8n Telegram Trigger – Messages and button callbacks
docs.n8n.io (Telegram Trigger)

Telegram Bot API – Buttons, messages, message deletion
core.telegram.org/bots/api

Cloudflare Tunnel – HTTPS access to the self-hosted server
developers.cloudflare.com (Tunnel)

Docker – Container deployment on the VPS
docs.docker.com

WhatsApp Cloud API (next) – Planned second channel
developers.facebook.com (WhatsApp)

Research and official sources

McAfee voice-cloning study (2023) – 3–4 s of audio gave ~85% voice match
ibtimes.com (14 Jun 2023)

MHA: ERSS 112 – India's single emergency number
mha.gov.in (ERSS)

National Cyber Crime Portal – Helpline 1930 for financial fraud
cybercrime.gov.in

Sanchar Saathi: Chakshu – Report suspected fraud calls
sancharsaathi.gov.in/sfc

Plivo docs: Calling in India – TRAI rule on India call media (why we avoid interception)
plivo.com/docs

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THANK YOU